Brightlearn Activity
Exercise 1: SQL Fundamentals: SELECT & Filtering

Section A: Selecting & Sorting

1. Query:

```
SELECT *
FROM employees;
```

Output:

id	first_name	last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. Query:

```
SELECT department
FROM
SELECT DISTINCT department
FROM employees;
```

Output

department
IT
HR
Finance
Marketing

3. Query:

```
SELECT first_name,
       last_name
FROM employees
ORDER BY salary DESC;
```

Output

First_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

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4. Query:

```
SELECT id,
       first_name,
       last_name,
       Salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

Output:

id	first_name	last_name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Section B: Filtering with WHERE

5. Query:

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

Output:

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Query:

```
SELECT id,
       First_name,
       last_name,
       department,
       Salary
FROM employees
WHERE department = 'Finance' AND salary > 60 000;
```

Output:

id	First_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000

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7. Query

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department='HR' OR department='Marketing';
```

Output

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Query:

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department NOT IN('IT');
```

Output:

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Query:

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department IN('IT','HR','Finance');
```

Output:

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

10. Query

```
SELECT id,
       first_name,
       last_name,
       department, Salary, City
FROM employees
WHERE department='IT' AND Salary>65 000 AND city='Johannesburg'
```

Output:

id	first_name	last_name	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg

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